

Solution Key

RC Circuit Derivations — Capacitor Charging Phase

1. a) Differential equation in u_C :

- Law of addition of voltages: $E = u_R + u_C$
- Ohm's Law: $u_R = R \cdot i$
- Current-charge relation: $i = \frac{dq}{dt}$
- Charge-voltage relation: $q = C \cdot u_C \implies i = C \frac{du_C}{dt}$
- Substitute i into u_R : $u_R = RC \frac{du_C}{dt}$
- Substitute u_R into the voltage law:

$$E = RC \frac{du_C}{dt} + u_C$$

b) Instantaneous capacitor charge:

- Using the direct proportional relationship: $q(t) = C \cdot u_C(t)$.
- (Optional note for marking: If u_C is expressed as $u_C(t) = E(1 - e^{-t/RC})$, then $q(t) = CE(1 - e^{-t/RC})$)

2. Differential equation in q :

- Law of addition of voltages: $E = u_R + u_C$
- Ohm's Law and current: $u_R = R \cdot i = R \frac{dq}{dt}$
- Voltage-charge relation: $u_C = \frac{q}{C}$
- Substitute u_R and u_C into the voltage law:

$$E = R \frac{dq}{dt} + \frac{q}{C}$$

3. a) Differential equation in i :

- Law of addition of voltages: $E = u_R + u_C$
- Differentiate the entire equation with respect to time t :

$$0 = \frac{du_R}{dt} + \frac{du_C}{dt}$$

- Differentiate $u_R = R \cdot i \implies \frac{du_R}{dt} = R \frac{di}{dt}$
- From $i = C \frac{du_C}{dt} \implies \frac{du_C}{dt} = \frac{i}{C}$
- Substitute back into the differentiated voltage law:

$$0 = R \frac{di}{dt} + \frac{i}{C}$$

b) Instantaneous voltage across the resistor:

- Using Ohm's Law: $u_R(t) = R \cdot i(t)$.
- (Optional note for marking: If $i(t)$ is expressed as $i(t) = \frac{E}{R}e^{-t/RC}$, then $u_R(t) = Ee^{-t/RC}$)

4. Differential equation in u_R :

- Start with the differential equation in i : $0 = R \frac{di}{dt} + \frac{i}{C}$
- From Ohm's law: $i = \frac{u_R}{R}$
- Differentiating gives: $\frac{di}{dt} = \frac{1}{R} \frac{du_R}{dt}$
- Substitute these into the equation for i :

$$0 = R \left(\frac{1}{R} \frac{du_R}{dt} \right) + \frac{1}{C} \left(\frac{u_R}{R} \right)$$

- Simplify to obtain:

$$0 = \frac{du_R}{dt} + \frac{u_R}{RC}$$

